
Sheet: to do

Загально-технічна документація

Android + iPhone
Expo SDK 57 / React Native 0.86

Версія застосунку: 0.5.0
Формат даних: AppData v2

Локальний offline-first застосунок без backend та облікового запису

Версія документа відповідає застосунку 0.5.0, Expo SDK 57 та React Native 0.86.

Паспорт проєкту

Аспект	Рішення
Тип	offline-first mobile planner
Платформи	Android та iPhone
Мови	українська й англійська
UI runtime	React Native 0.86, React 19
Platform layer	Expo SDK 57, CNG, EAS/Xcode/Gradle
Навігація	Expo Router, file-based routes
Стан	React Context + synchronous dataRef
Зберігання	AsyncStorage і приватні Expo FileSystem files
Сповіщення	Notifee exact alarm / Time Sensitive notification
Мережа	не потрібна для роботи застосунку
Backend/API	відсутні

Мета та межі

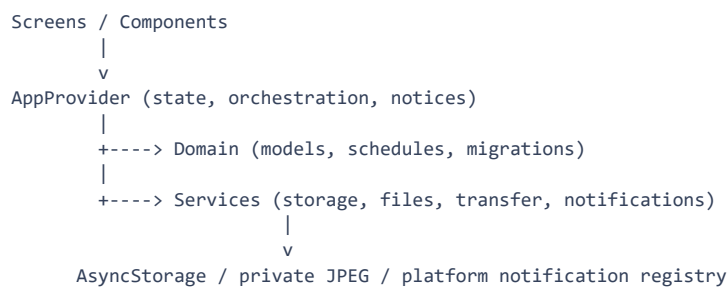
Sheet: to do дає змогу вести незалежні списки, планувати їх у календарі, прикріплювати фотографії й отримувати локальні reminders без облікового запису, сервера або cloud sync. Один installation sandbox є одним локальним профілем.

Платформи підтримуються в одному Expo-проєкті. Generated android/ та ios/ відтворюються з app.json і не є джерелом істини в Git.

Функціональні модулі

Модуль	Можливості	Реалізація
Списки	CRUD, drag-and-drop, progress	ListCard, domain functions
Завдання	CRUD, completion, task photos	TaskRow, AppProvider
Календар	month grid та agenda	CalendarGrid, /calendar
Reminders	list/task local schedule	Notifee, /alarm
Теми	чотири фони та custom JPEG	theme registry, file store
Transfer	text, JSON/deep link, .sheettodo	transfer/package services
Import	picker, file association, deep link	/import-file, /import
Локалізація	uk/en, calendar locale	translations and locale helpers

Архітектурна позиція



Рівень	Відповідальність
UI	input, confirmation, accessibility, responsive layout
State	orchestration, busy state, persistence calls, cleanup
Domain	schema, limits, validation, immutable transitions
Services	storage, files, archive parsing, sharing, notifications
Platform config	identifiers, permissions, UTI/MIME, signing profiles

Business logic, models, state, storage, transfer, theme tokens and components are shared. Platform branching is limited to notification options, permissions and native build configuration.

Маршрути

Route	Призначення	Параметри
/	lists and tasks	—
/calendar	calendar and agenda	—
/alarm	active alarm screen	notification ID, title, body
/import	deep-link confirmation	encoded data
/import-file	.sheettodo confirmation	encoded file uri

+native-intent.tsx converts Android content:// and Android/iOS file:// URI into /import-file. Scheme sheettodo handles compact JSON deep links. On iOS, the exported UTI com.lilydotsenko.sheettodo.list associates .sheettodo with the app.

Expo Router provides native stacks. Safe areas are applied to every full screen, Android predictive back stays enabled and iOS uses the native edge-back gesture.

Життєвий цикл даних

Запуск

1. `AppStorage.load()` reads JSON from `AsyncStorage`.
2. `migrateStoredData()` validates or upgrades the stored shape.
3. `scrubMissingFiles()` checks task photos and custom background.
4. Clean data becomes current `dataRef` and React state.
5. Invalid raw payload is copied to the corrupt backup key and reset safely.

Зміна

1. UI invokes an `AppProvider` operation.

2. A domain function returns a new `AppData` value.
3. `commit()` synchronously updates `dataRef` and React state.
4. Serialized snapshots enter the `AsyncStorage` write queue.
5. A failed write produces `saveError` without crashing the UI.

The write queue prevents a slower older snapshot from overwriting a newer one.

Списки та завдання

- up to 100 lists and 500 tasks per list;
- list title: 1–60 characters;
- task text: 1–160 characters;
- progress is derived from completed tasks and is not stored separately;
- completing a task cancels and clears its reminder;
- unknown list/task IDs result in a safe no-op;
- imported aggregates always receive new IDs.

Notifications

The stored schedule shape is platform-independent: `scheduledAt`, `alarmEnabled` and `notificationId`. ID values follow `list-{id}` or `task-{id}`.

Behavior	Android	iOS
Trigger	exact timestamp + <code>AlarmManager</code>	timestamp local notification
Importance	alarm channel, high	Time Sensitive
Sound	default, looping	default system sound
Presentation	ongoing + full-screen action	banner/list notification
Stop	notification action and alarm screen	notification action and alarm screen
Extra permission	exact alarms on Android 12+	none beyond notifications

`configureNotifications()` creates the Android channel or iOS action category. `scheduleReminder()` validates future time and requests platform permission. The Android-only `AlarmManager` object is never passed to the iOS trigger.

On Android, `getInitialNotification()` handles cold-start navigation. On iOS, `Notifee PRESS` events open `/alarm`; `DELIVERED` opens the screen only while the application is in foreground. Background stop actions run from the early handler registered in `index.ts`.

iOS intentionally does not request Critical Alert entitlement. It cannot match Android full-screen/looping behavior because that capability is restricted by the operating system.

`expo-notifications` remains a compatibility dependency only for cancelling old Android notification IDs produced by version 0.3. New reminders use `Notifee`.

Images and themes

`ImagePicker` gives access only to user-selected media. Selected content is resized, compressed as JPEG and copied into the private document directory.

Type	Directory	Maximum edge	JPEG quality
Task photo	task-photos	1600 px	0.76
Custom background	custom-backgrounds	2400 px	0.82

Deletion and replacement clean managed files. Startup removes metadata references to missing files. If a task disappears while the picker is open, the newly copied file is removed instead of becoming an orphan.

Import and export

Text share is human-readable. `.sheettodo` is a ZIP containing `manifest.json` and JPEG files under `photos/`. It transfers one list, task completion state and available photos; schedules, notification IDs, creation timestamps and custom background are not exported.

The parser validates format, version, archive size, entry count, path safety, duplicates, JPEG markers, declared sizes and CRC32. Legacy web JSON, older ZIP packages and compact deep links remain readable.

Temporary export files are created under cache and deleted in `finally`. A package photo import is compensating: if a later write fails, earlier imported files are removed.

UX and adaptive layout

- safe-area containers protect notches and home indicator;
- `KeyboardAvoidingView` uses iOS padding and Android resized height;
- focused task inputs are measured and scrolled above the keyboard;
- one list column is used below 760 px and two above it;
- cards and dialogs have bounded widths for large screens;
- stable accessibility labels, roles and E2E `testID` values are present;
- the app keeps a light system shell and offers four visual background themes.

System dark mode is not introduced in 0.5.0 because the Android behavior being preserved did not provide it.

Security and privacy

- no backend, analytics, remote API, auth or tokens;
- no required environment variables;
- no app-level encryption for `AsyncStorage`;
- import is treated as untrusted input;
- ATS disallows arbitrary HTTP; local networking remains available for dev builds;
- camera, microphone and geolocation are not requested;
- push entitlement is removed because only local notifications are used;
- signing keys, certificates and provisioning profiles are ignored by Git.

Data leaves the device only after the user invokes Share/Export. Uninstall removes the private app sandbox.

Build configuration

Profile	Android	iOS	Purpose
development	internal APK	signed device development build	native debugging
development-simulator	—	Simulator development build	unsigned Simulator QA
preview	installable APK	ad hoc IPA	personal/internal installation
production	AAB	store-signed archive	retained, not submitted

Application version is 0.5.0; Android package and iOS bundle identifier are `com.lilydotsenko.sheettodo`. Android `versionCode` and iOS `buildNumber` are 5. No submit profile or automated store publication exists.

Quality control

Check	Command
Formatting	<code>npm run format:check</code>
Lint	<code>npm run lint</code>
TypeScript	<code>npm run typecheck</code>
Unit tests	<code>npm test</code>
Integration	<code>npm run test:integration</code>
Expo compatibility	<code>npm run doctor</code>
Android bundle	<code>npx expo export --platform android</code>
iOS bundle	<code>npx expo export --platform ios</code>
Device E2E	<code>npm run test:e2e</code>

The repository contains 43 Jest tests in 13 suites, including notification configuration, iOS URI routing and an offline storage/export/import restart flow. Maestro covers core UI persistence, navigation/theme and deep-link import.

Native permissions, photo picker, OS share sheets, exact alarm and Time Sensitive delivery require the physical-device matrix in `.maestro/DEVICE_CHECKLIST.md`.

GitHub Actions runs non-signing checks and both platform exports. It does not build or publish store artifacts and does not access signing credentials.

Directory map

Path	Role
<code>src/app</code>	screens and routes
<code>src/components</code>	reusable UI
<code>src/domain</code>	models, schedule helpers, migrations
<code>src/services</code>	storage, files, transfer, notifications
<code>src/state</code>	application state orchestration
<code>src/theme</code>	themes and icon registry
<code>plugins</code>	reproducible native configuration
<code>.maestro</code>	E2E flows and device checklist
<code>docs</code>	install, technical and data documents

Known limitations

- no cloud sync or multi-device merge;
- `.sheettodo` transfers one list per file;
- custom backgrounds are not included in list backup;
- local notification delivery depends on OS settings;
- iOS Personal Team builds expire after 7 days;
- a native iPhone build requires macOS/Xcode or paid EAS signing;
- direct file opening depends on the source application's MIME/UTI behavior.

Conclusion

Version 0.5.0 extends the existing Android application to iPhone without a framework rewrite. Shared TypeScript remains the source of product behavior, while platform-specific configuration is limited, explicit and reproducible.